实验四.摄像头操作

实验目的

1. 巩固 ROS2 指令操作
2. 了解图像格式（jpeg、nv12）和数据传送方式（ros、share_mem）

usb 摄像头型号及参数



感光芯片：OV2710
最大分辨率：1920×1080（200 万像素）
光圈（F）：2.8
焦距（EFL）：2.9mm
视场角（FOV）：145°
支持图像格式：MJPEG、YUY2
接口类型：USB2.0

实验过程

1. 启动摄像头（jpeg 格式）

1）第一个窗口，输入

ros2 launch originbot_bringup camera.launch.py　　**只启动摄像头**

或

ros2 launch originbot_bringup originbot.launch.py use_camera:=true

启动底盘通讯和摄像头

```
root@ubuntu:~# ros2 launch originbot_bringup camera.launch.py
[INFO] [launch]: All log files can be found below /root/.ros/log/2025-03-22-20-36-13-679414-ubuntu-4735
[INFO] [launch]: Default logging verbosity is set to INFO
config_file_path is  /opt/tros/humble/lib/hobot_usb_cam/config/usb_camera_calibration.yaml
Hobot shm pkg enables zero-copy with fastrtps profiles file: /opt/tros/humble/lib/hobot_shm/config/shm_fastdds.xml
Hobot shm pkg sets RMW_FASTRTPS_USE_QOS_FROM_XML: 1
env of RMW_FASTRTPS_USE_QOS_FROM_XML is  1 , ignore env setting
[INFO] [hobot_usb_cam-1]: process started with pid [4736]
[hobot_usb_cam-1] [WARN] [1742646974.113655733] [hobot_usb_cam]: framerate: 30
[hobot_usb_cam-1] [WARN] [1742646974.114190362] [hobot_usb_cam]: pixel_format_name: mjpeg
[hobot_usb_cam-1] [WARN] [1742646974.135076248] [hobot_usb_cam]: Camera calibration file: [/opt/tros/humble/lib/hobot_usb_cam/config/usb_camera_calibration.yaml] does not exist!
[hobot_usb_cam-1] If you need calibration msg, please make sure the calibration file path is correct and the calibration file exists!
[hobot_usb_cam-1] [WARN] [1742646974.145936049] [hobot_usb_cam]: Try to open device [/dev/video0]
[hobot_usb_cam-1] [WARN] [1742646974.481363108] [hobot_usb_cam]: This devices supproted formats:
[hobot_usb_cam-1] [WARN] [1742646974.481969654] [hobot_usb_cam]:        Motion-JPEG: 1920 x 1080 (30 Hz)
[hobot_usb_cam-1] [WARN] [1742646974.482166114] [hobot_usb_cam]:        Motion-JPEG: 720 x 1000 (30 Hz)
[hobot_usb_cam-1] [WARN] [1742646974.482333699] [hobot_usb_cam]:        Motion-JPEG: 640 x 480 (30 Hz)
[hobot_usb_cam-1] [WARN] [1742646974.482499159] [hobot_usb_cam]:        Motion-JPEG: 1280 x 720 (30 Hz)
[hobot_usb_cam-1] [WARN] [1742646974.482667244] [hobot_usb_cam]:        Motion-JPEG: 640 x 360 (30 Hz)
[hobot_usb_cam-1] [WARN] [1742646974.482830495] [hobot_usb_cam]:        Motion-JPEG: 320 x 240 (30 Hz)
[hobot_usb_cam-1] [WARN] [1742646974.482993580] [hobot_usb_cam]:        Motion-JPEG: 800 x 600 (30 Hz)
[hobot_usb_cam-1] [WARN] [1742646974.483156581] [hobot_usb_cam]:        Motion-JPEG: 600 x 600 (30 Hz)
[hobot_usb_cam-1] [WARN] [1742646974.483318958] [hobot_usb_cam]:        YUYV 4:2:2: 1920 x 1080 (5 Hz)
[hobot_usb_cam-1] [WARN] [1742646974.483479501] [hobot_usb_cam]:        YUYV 4:2:2: 720 x 1000 (5 Hz)
[hobot_usb_cam-1] [WARN] [1742646974.483641085] [hobot_usb_cam]:        YUYV 4:2:2: 640 x 480 (30 Hz)
[hobot_usb_cam-1] [WARN] [1742646974.483802962] [hobot_usb_cam]:        YUYV 4:2:2: 1280 x 720 (10 Hz)
[hobot_usb_cam-1] [WARN] [1742646974.483964505] [hobot_usb_cam]:        YUYV 4:2:2: 640 x 360 (15 Hz)
[hobot_usb_cam-1] [WARN] [1742646974.484126673] [hobot_usb_cam]:        YUYV 4:2:2: 320 x 240 (15 Hz)
[hobot_usb_cam-1] [WARN] [1742646974.484287799] [hobot_usb_cam]:        YUYV 4:2:2: 800 x 600 (10 Hz)
[hobot_usb_cam-1] [WARN] [1742646974.484450217] [hobot_usb_cam]:        YUYV 4:2:2: 600 x 600 (10 Hz)
[hobot_usb_cam-1] [WARN] [1742646974.557076334] [hobot_usb_cam]: Open & Init device /dev/video0 success.
[hobot_usb_cam-1] [WARN] [1742646974.617582140] [hobot_usb_cam]: Enabling zero-copy
```

2）第二个窗口中输入

ros2 topic list

```
root@ubuntu:~# ros2 topic list
/camera_info
/hbmem_img
/image
/parameter_events
/rosout
root@ubuntu:~#
```

ros2 topic type /image

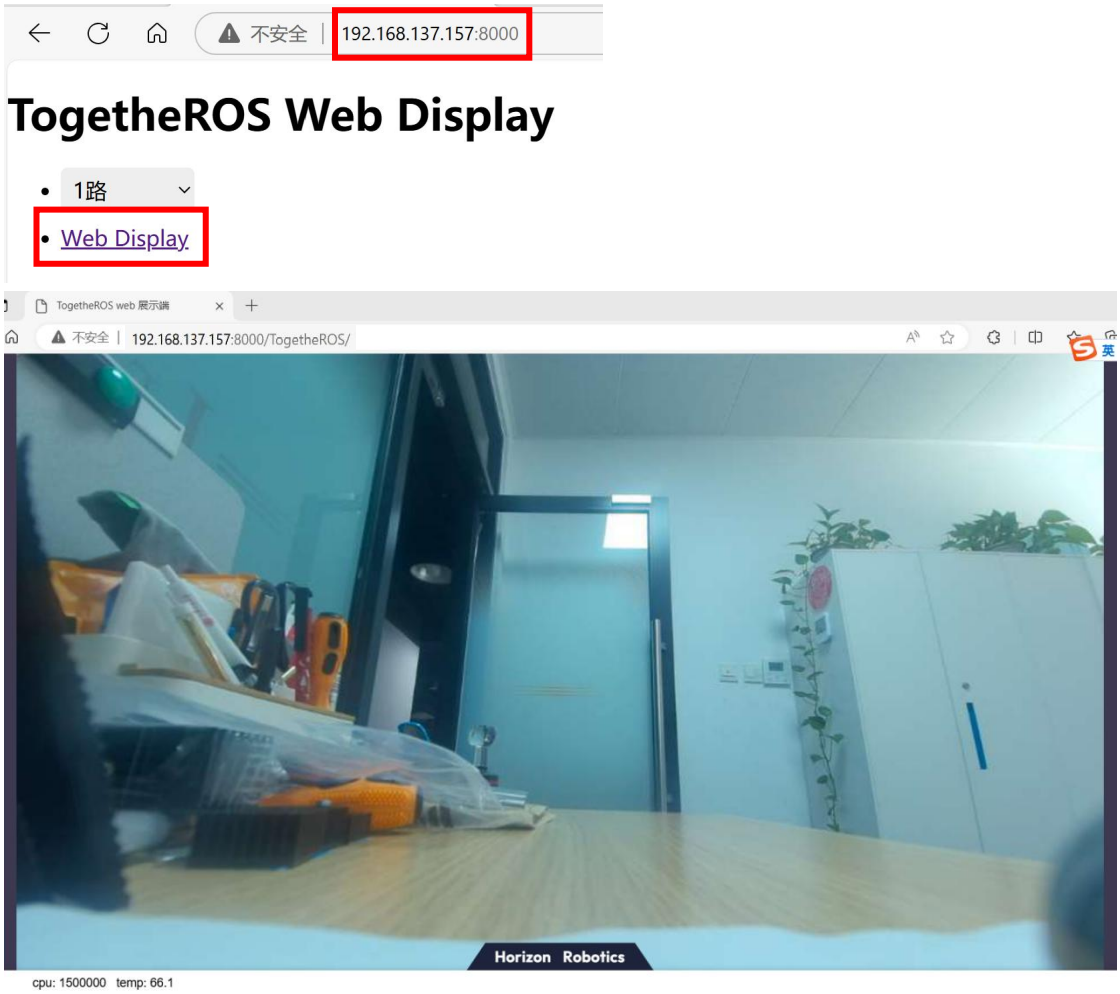
```
root@ubuntu:~# ros2 topic type /image
sensor_msgs/msg/CompressedImage
root@ubuntu:~#
```

ros2 topic type /hbmem_img

```
root@ubuntu:~# ros2 topic type /hbmem_img
hbm_img_msgs/msg/HbmMsg1080P
root@ubuntu:~#
```

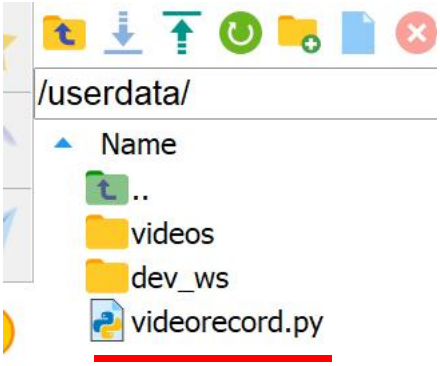
2. 网页显示图像

在浏览器（IE）中，输入小车 ip:8000，点击 Web 显示端，就可以查看图像了。



3. 视频录制

上传 videorecord.py 程序到/userdata 目录下



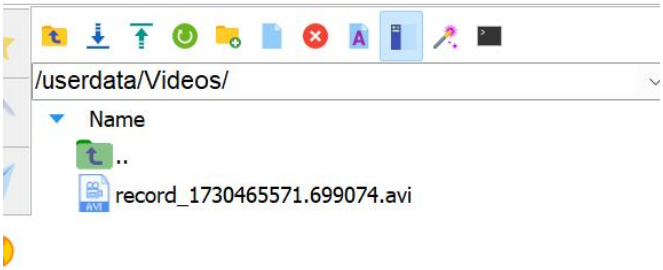
在第二个窗口，输入两条命令：

cd /userdata

python videorecord.py

```
root@ubuntu:~# cd /userdata/
root@ubuntu:/userdata# python videorecord.py
[INFO] [1730467815.528777991] [video_recorder]: Video Recording.
```

录制的视频文件在/userdata/videos 目录下，按 ctr+c 中断程序完成视频录制，视频可下载到本地计算机后再打开。



4. 拍照

将 picker_pic.py 文件上传到/userdata 目录下

在第二个窗口，启动图像采集程序，运行

cd /userdate/

python picker_pic.py

每按下一次回车键，采集一张图像，

```
root@ubuntu:~# cd /userdata/
root@ubuntu:/userdata# python3 picker_pic.py
[INFO] [1730719417.973353405] [picture_recorder]: start picking picture!
record 0 picturs
record 1 picturs
record 2 picturs
record 3 picturs
record 4 picturs
record 5 picturs
```

采集结束后，多按几下 ctr+c 终止程序。

图像数据存在/userdata/pictures/img 文件夹下，如果没有img文件夹，自己新建一下。

附 videorecord.py 程序

```
#!/usr/bin/env python
# -*- coding: utf-8 -*-

import rclpy, cv2, cv_bridge
from rclpy.node import Node
from sensor_msgs.msg import  CompressedImage
import time
import numpy as np
from cv_bridge import CvBridge

W = 1920
H = 1080

class VideoRecord(Node):
    def __init__(self):
        super().__init__('video_recorder')
        self.get_logger().info("Video Recording.")
        self.bridge = CvBridge()
        self.out = cv2.VideoWriter('/userdata/videos/record_' + str(time.time()) + '.avi', cv2.VideoWriter_fourcc('X','V','I','D'), 5, (W, H))

        self.image_sub = self.create_subscription(CompressedImage, '/image', self.image_callback, 10)

    def image_callback(self, msg):
        self.image = CvBridge().compressed_imgmsg_to_cv2(msg)
        self.out.write(self.image)

def main(args=None):
    rclpy.init(args=args)
    videorecord = VideoRecord()
    rclpy.spin(videorecord)
    videorecord.destroy_node()
    rclpy.shutdown()

if __name__ == '__main__':
    main()
```